



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. 8390

AVI: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 305	Seat No: 2022700054
Course Section:1		

L_41_667_986_L_1505_41373

Name of the candidate. Ritvik Sharma

Candidate's UID number:

2	0	2	2	7	0	0	0	5	4
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Examination class room number:

3	0	5
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Invigilator's signature with date:

	8/10
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(To be filled in by the candidate)

EXAMINATION: BE BTech
(For example FY MCA/FY BTECH)

PROGRAM: CSE-DS

DAY AND DATE: 8th October 2025, Wednesday

COURSE NAME:

SEMESTER : I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)								
Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Q19	<u>Solution:-</u> We have $f(w) = \frac{1}{2} \ Xw - y\ ^2$ Assume this as the error function we can write $L(w) = \frac{1}{2} \ Xw - y\ ^2$ $\therefore \nabla L(w) = \ Xw - y\$ Suppose w^* is the optimal solution for the above equation. By Taylor series we can write $h = w - w^*$ $L(w^* + h) = L(w^*) + h \nabla L(w^*) + H(w - w^*)$ at w^* we have $\nabla L(w^*) \rightarrow 0$ $\therefore L(w) = H(w - w^*)$ By singular value decomposition we can write $H = Q \Lambda Q^T$ So $L(w) = Q \Lambda Q^T (w - w^*)$ By definition we have $Xw = h$ $\therefore w = X^+ h$ $L(w) = Q \Lambda Q^T (X^+ h - w^*)$ $= Q \Lambda Q^T (X^+) (\mathbb{I} h - w^*)$ where $\mathbb{I}$ is an identity matrix substitute $Q = P \text{ and } Q^T X^+ = P^T$ by the equivalence of Q and P where P is a matrix with columns as eigenvalues of $X^T X$ $\therefore L(w) = P \Lambda P^T (\mathbb{I} h - w^*)$ this is of the form $U \Sigma V^T$ where $\Sigma = \Lambda$ and Λ is a matrix of eigenvalues of $X^T X$

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$$\Delta(W) = P \begin{bmatrix} \frac{1}{\lambda_1} & & & \\ & \ddots & & \\ & & \ddots & \\ & & & \frac{1}{\lambda_n} \end{bmatrix} P^T$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues from most dominant to least dominant. The maximum value will occur when $\|\Sigma\| < \frac{1}{\lambda_{\max}}$

$$\text{as } W_{t+1} = W_t - \eta \Delta(W)$$

and $\|\Delta(W)\| \leq \frac{1}{\lambda_{\max}}$ we see that

$$\eta < \frac{1}{\lambda_{\max}} \Rightarrow \eta < \frac{2}{\lambda_{\max}} \quad \text{where}$$

$\lambda_{\max}$ is largest eigenvalue (dominant) of $X^T X$

Hence proved

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Q1b

Solution :-

Suppose we have n training points
and m testing points

The loss function we assume is the
MSE $\Rightarrow \mathcal{L}(\theta) = \sum_{i=1}^n \sum_{j=1}^m (\hat{y}_{ij} - y_{ij})^2$

This can be written in expectation
as $E[(\hat{f}(x_i) - f(x_i))^2]$

we compute this empirically as
we know $\hat{y}$ and y
suppose

$$\hat{y} = \hat{f}(x) \text{ and}$$

$$y = f(x) + \epsilon \text{ where } \epsilon \text{ is an irreducible error}$$

$$\begin{aligned} E[(\hat{y} - y)^2] &= E[(\hat{f}(x) - f(x) - \epsilon)^2] \\ &= E[(\hat{f}(x) - f(x))^2 - 2\epsilon(\hat{f}(x) - f(x)) + \epsilon^2] \\ &= E[(\hat{f}(x) - f(x))^2] - 2E[(\epsilon(\hat{f}(x) - f(x)))] + E[(\epsilon^2)] \end{aligned}$$

On rearranging we get

$$E[(\hat{f}(x) - f(x))^2] = E[(\hat{y} - y)^2] + 2E[\epsilon(\hat{f}(x) - f(x))] - E[(\epsilon^2)]$$

From the definition of loss function

$$E[(\hat{y} - y)^2] = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

$$E[(\epsilon^2)] = \frac{1}{n} \sum_{i=1}^n \epsilon_i^2$$

and $E[\epsilon(\hat{f}(x) - f(x))]$ is the
covariance between ϵ and
 $\hat{f}(x) - f(x)$

we can say this as

true error = empirical error - irreducible
error + model complexity

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By Stein's lemma we have

$$E[(\hat{f}(x) - f(x))^2] = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 + \frac{1}{n} \sum_{i=1}^n \epsilon_i^2 - \frac{\sigma^2}{n} \sum_{i=1}^n \frac{\partial \hat{f}(x)}{\partial \hat{y}_i}$$

Hence as model complexity increases we get a false picture of how close $\hat{f}(x)$ is to $f(x)$. Hence it will affect generalization of model and we will have to rely on validation error



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Q2a

Solution:-

Consider the following parameters

Input volume = $(H \times W \times C)$ Filter volume = $(F \times F)$ we have K filters

we assume that we do not apply any padding to the output volume. Reduces by a quantity $(F-1)$. Also assume stride $S=1$ for simplicity

output volume = $(H - (F-1), W - (F-1), K)$

If we were to include padding and stride then after adjusting for the floor and ceil we get

$$H_2 = \frac{H_1 - F + 2P + 1}{S}$$

$$W_2 = \frac{W_1 - F + 2P + 1}{S}$$

$$D_2 = K$$

The number of learnable parameters are the weights of filters and biases. The parameters are

$$= F \times F \times C \times K + \underbrace{K}_{\text{biases}}$$

Hence the time complexity of one hidden layer is $O(F^2 C K)$

where $F \rightarrow$ filter size

$C \rightarrow$ input depth

$K \rightarrow$ number of filters

In depthwise separable convolutions we have a separate connection between the convolutional layers

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The idea is to train the convolutional layer with a number of filters over the entire image and replace the learned weights with the other layers. This is in accordance to dropout mechanism where one neuron is trained at the probability of $\frac{1}{2^n}$

but guarantees the overall training of all neurons. This will reduce the computational complexity in the convolutional layer from the factor of F^2 and will result in efficient learning of the network.

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Q2b

Solution:-

We make a claim such that the autoencoder has

- ① Linear encoder
- ② Linear decoder
- ③ MSE as the error function
- ④ Input normalised to

$$\hat{x}_{ij} = \frac{1}{\sqrt{m}} \left(x_{ij} - \sum_{i=1}^m x_{ij} \right)$$

It is sufficient for us to show that if the decoder is linear and MSE function is used, then it requires the encoder to be linear. The transformation is essentially normalization as

$$X' = \frac{1}{\sqrt{m}} X$$

$$\begin{aligned} \therefore (X')^T X' &= \frac{1}{\sqrt{m}} \times \frac{1}{\sqrt{m}} X^T X \\ &= \frac{1}{m} X^T X \end{aligned}$$

hence we are essentially finding the covariance matrix.

The output of the decoder will be $\min(\|HW^* - y\|_F^2)$

where $\|A\|_F^2$ is the Frobenius norm. By SVD we know that $HW^* = U_{K \times K} \Sigma_{K \times K} V_{K \times K}^T$

$$\text{assume } H = U_{K \times K} \Sigma_{K \times K}$$

$$\begin{aligned} \therefore H &= (X X^T) (X X^T)^+ U_{K \times K} \Sigma_{K \times K} \\ &= X (U \Sigma V^T) (U \Sigma V^T (U \Sigma V^T)^+)^+ \\ &\quad U_{K \times K} \Sigma_{K \times K} \end{aligned}$$

$$\text{as } X = U \Sigma V^T$$

$$(ABC)^T = C^T B^T A^T$$

$$(ABC)^T = C^T \otimes B^T A^T$$

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$$H = (X U \Sigma V^T) (U \Sigma V^T V \Sigma^T U^T)^{-1} U_{\leq K} \Sigma_{K \times K}$$

we have $V^T V = I$ $U U^T = I$
 $\Sigma \Sigma^T = I$

$I \rightarrow$ Identity matrix

hence essentially we would end up with

$$(U \Sigma V^T V \Sigma^T U^T)^{-1} = I$$

$$\begin{aligned} \therefore H &= (X U \Sigma V^T) I U_{\leq K} \Sigma_{K \times K} \\ &= X U \Sigma V^T U_{\leq K} \cdot \Sigma_{K \times K} \\ &= X U_{\leq K} \Sigma_{K \times K} V^T \end{aligned}$$

$$= X V_{\leq K} \quad \text{---} \quad (U_{\leq K} \cdot \Sigma_{K \times K} = I)$$

Hence H is a linear function of X and V comes from the eigenvectors of $X^T X$. Hence we can say that linear undercomplete autoencoder is the same as PCA

An over complete autoencoder has $\dim(h) \geq \dim(x)$. Hence the encoder tends to copy the input x_i to h_i and then again copies h_i towards $\hat{x}_i$. Hence we say that over complete autoencoder tends to learn the identity function



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Q3a Solution :-

In an RNN we can write the expression for the loss function as the cross entropy function

$$L(\theta) = - \sum_{i=1}^M \sum_{j=1}^J \left[x_{ij}^{(0)} \log \hat{x}_{ij}^{(0)} + (1 - x_{ij}^{(0)}) \log (1 - \hat{x}_{ij}^{(0)}) \right]$$

This expression is only minimized when $x_{ij}^{(0)} = \hat{x}_{ij}^{(0)} = 0$ or 1

By chain rule we can write

$$\frac{\partial L(\theta)}{\partial h_i} = \frac{\partial L(\theta)}{\partial x_{ij}} \cdot \frac{\partial x_{ij}}{\partial h_{ij}}$$

we know that x_{ij} and h_{ij} are related by sigmoid function i.e

$$h_{ij} = \sigma(w_i^T x_i + b)$$

$$\therefore \frac{\partial h_{ij}}{\partial x_{ij}} = \sigma(f(x)) \cdot (1 - \sigma(f(x)))$$

$$\frac{\partial L(\theta)}{\partial \hat{x}_{ij}^{(0)}} = \frac{\partial}{\partial \hat{x}_{ij}^{(0)}} \left[- \sum_{i=1}^M \sum_{j=1}^J x_{ij}^{(0)} \log \hat{x}_{ij}^{(0)} + (1 - x_{ij}^{(0)}) \log (1 - \hat{x}_{ij}^{(0)}) \right]$$

$$= - \frac{x_{ij}^{(0)}}{\hat{x}_{ij}^{(0)}} + \frac{1 - x_{ij}^{(0)}}{1 - \hat{x}_{ij}^{(0)}}$$

Hence we can say

$$\frac{\partial L}{\partial h_{t-k}} = \left(- \frac{x_{kj}^{(0)}}{\hat{x}_{kj}^{(0)}} + \frac{1 - x_{kj}^{(0)}}{1 - \hat{x}_{kj}^{(0)}} \right) \sigma(f(x)) \cdot (1 - \sigma(f(x)))$$

The gradient expression has a component of the learning rate and weights (η and w) since RNN uses loops to remember the last information ~~and~~ the exponentiation of η and w tends

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to explode when $(\eta > 1)$ and
vanishes when $(\eta < 1)$

The forget gate of LSTM determines what information should be remembered and what has to be forgotten over time. The expression is

$$f_t = f(h(t-1)W(t) + b_f)$$

where f is generally the logistic function and determines the output to be either 0 or 1

$$\text{logistic} = \frac{1}{1 + e^{-z}}$$

when $z \rightarrow \infty$ logistic $\rightarrow 0$

$z \rightarrow -\infty$ logistic $\rightarrow 1$

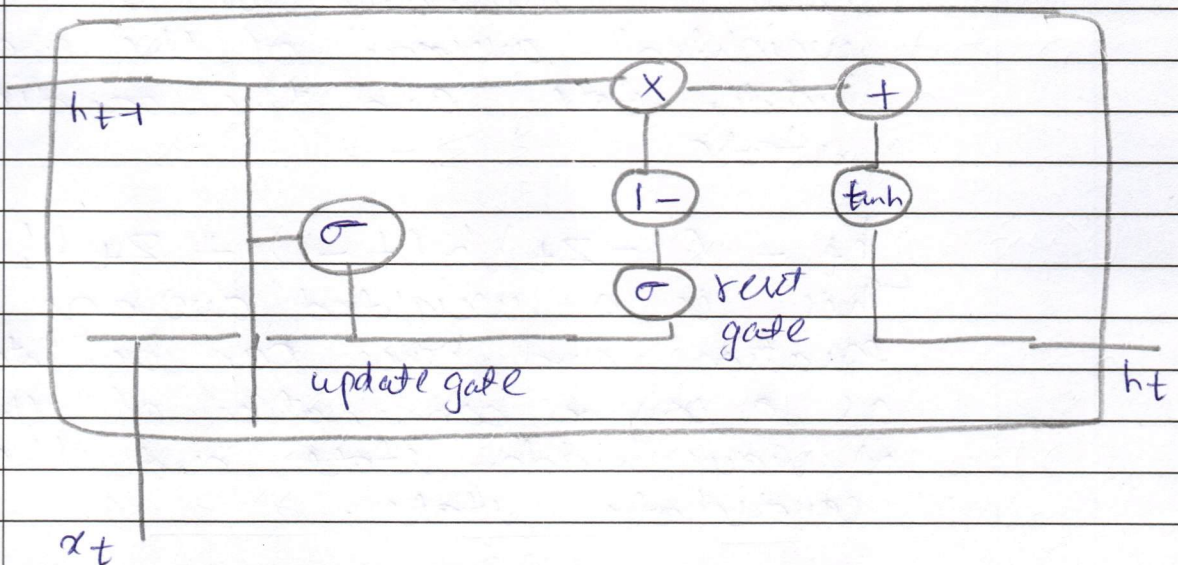
$z \rightarrow 0$ logistic $\rightarrow 0.5$

because we do not have a factor of η on the numerator the problem of exploding and vanishing gradients is solved when the output of logistic function is 0, the information is forgotten and when output is 1 the information is retained in the cell state of LSTM



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Q3b Solution :-



The GRU contains of two gates the update gate and reset gate
 ① The update gate determines what information has to be retained from the previous state

$$z_t = f(h(t-1), x(t), w(t), b)$$

where f is generally a sigmoid function

② The reset gate determines what information has to be forgotten

$$r_t = f(w_r(t), h(t-1), x(t), b)$$

where f is again a sigmoid function

③ The candidate hidden state is calculated by using the tanh function and the output of the reset gate

$$h'_t = \tanh[r_t w_i(t) \cdot h(t-1) x(t)]$$

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- ④ The potential hidden state can now be calculated as the weighted average of the previous hidden state and the candidate state

$$h_t = (1 - z_t) h(t-1) + z_t h'_t$$

This is a weighted average equation and hence can be interpreted as a convex combination of the previous hidden state and the candidate state.

- ⑤ The GRUs have combined the forget and input gate of LSTMs to a single update gate and determine what information has to be forgotten or retained in a single gate. This leads to reduction of parameters

For LSTM

$$f_t = f(h(t-1), w_f(t), x(t))$$

$$i_t = f(h(t-1), w_i(t), x(t))$$

$$o_t = \tanh(h(t-1), w_o(t), x(t))$$

These three equations have been reduced to

$$z_t = f(h(t-1), x(t), w(t), + b)$$

hence the parameters are reduced

- ⑥ They can be used in language translation, image captioning and other tasks which require high GPU computation but are efficient on GRU due to reduced parameters

[illegible]

[illegible]